

Written Report

What characteristics are associated with depression and anxiety? We are curious to determine what trends may appear throughout Wachusett High School's student body regarding mental health and personal life. Our goal is not to find evidence regarding which characteristics or if the number of characteristics may cause these health concerns but rather which responses show a pattern of association.

We conducted a survey to gather information on the high school students within the Wachusett population. Since we collected primarily categorical data that required direct questioning, a survey worked best as a method rather than an observational study or an experiment. Also, the latter data collection methods required information already collected or tested, which could not have been done with the type of data we needed. The contents of our survey were organized using an online Google Form provided to the sample via school email. It also included an explanation section, a statement that results were to be anonymous, and the context of our survey project. If students failed to respond within a week, we sent a follow-up email encouraging them to participate in our survey, which avoided potential nonresponse bias. After two weeks had passed, our survey closed.

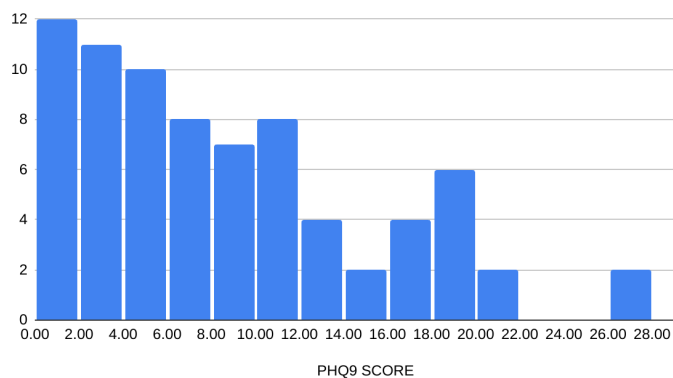
The following questions were asked: student grade ("What is your grade level?"), gender ("What is your gender?"), daily caffeine intake ("What is your daily caffeine intake"), music genres regularly listened to ("What music genres do you regularly listen to?"), sports team membership ("What sports team have you played in?"), Patient Health Questionnaire-9, receiving treatment for any form of depression, General Anxiety Disorder-7, and receiving treatment for any form of anxiety. All questions besides grade and gender were prompted by multiple-choice responses, which were required to answer, and respondents could choose more than one response. The question about daily caffeine intake had answers that qualitatively categorized caffeine intake with responses such as one cup of coffee, one energy drink, etc., because some people may not have known the exact milligrams of caffeine they consume daily. We sorted each response type by number and type of drink. We asked about regular music genres listened to, providing a list of popular genres grouped together by similarity, an option for none, and an option for "other" where respondents could type a different genre that was not on our list. The sports membership question allowed respondents to select as many sports teams as they belonged to, as we had provided a lengthy list of common athletic teams to choose from and again options for none or other. Medical professionals wrote and determined the two health-related questions (Patient Health Questionnaire-9 and General Anxiety Disorder-7) to accurately test mental health symptoms and behaviors according to their standards. This avoided potential response bias as professionals have agreed upon what is acceptable to ask. Respondents reported the frequency of a list of symptoms over the past two weeks for these questions. Each frequency corresponded to a number from zero (Not at All) to three (Nearly Every Day). The

summed final scores were then sorted into categories from mild to severe. Respondents then answered with their final score in our survey question, as the medical questions were provided on a separate website.

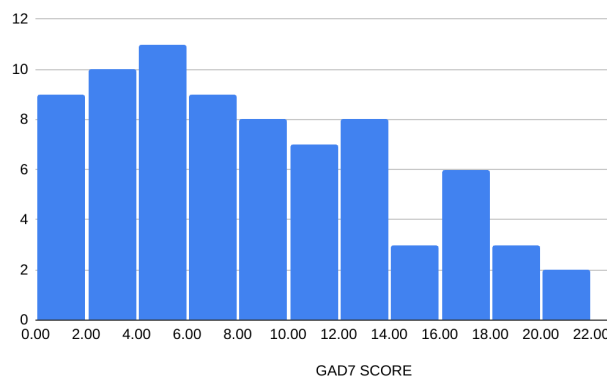
To randomize, our sampling procedure was a stratified random sample. We obtained our sampling frame (a list of all Wachusett students, not including post-grads), stratified them by the four grade levels (9th, 10th, 11th, and 12th), and conducted a simple random sample. Then, we started with 9th grade and assigned each student a unique number. Using a random number generator, discarding repeated numbers and numbers over the greatest student number, we selected 50 students. This process was repeated for each grade. We considered there may be different trends in mental health in the four different grades, so we stratified the students to reduce variability.

Once data were collected from students, it was analyzed and put into charts for visualization; due to the large amount of qualitative data, these primarily consisted of side-by-side bar charts with mean comparisons. In addition, we inputted the data into box plots, calculating and comparing the medians and IQRs. We then determined the association between the characteristics and depression and anxiety, using separate bar charts for each characteristic and comparing them across grade levels. Considering possible lurking variables and other sources of bias, we used the information collected to conclude potential associations between our chosen personal characteristics and mental health conditions.

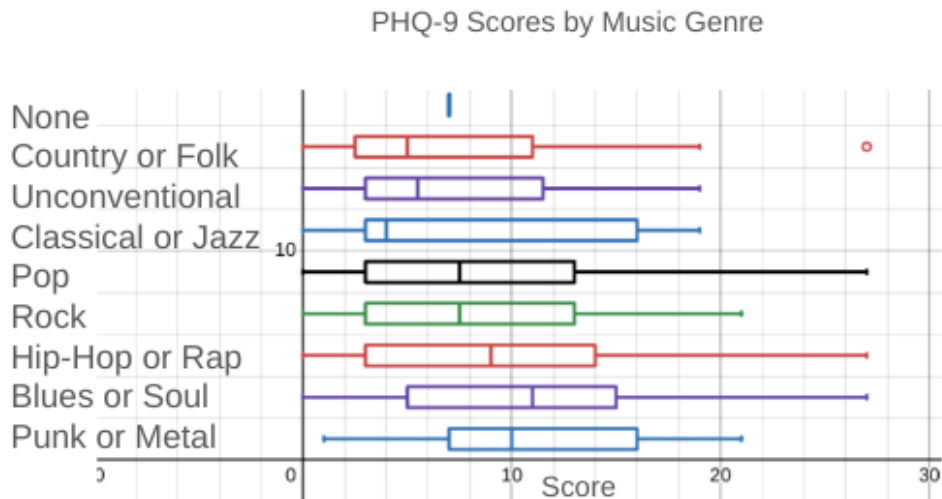
Histogram of PHQ9 SCORE



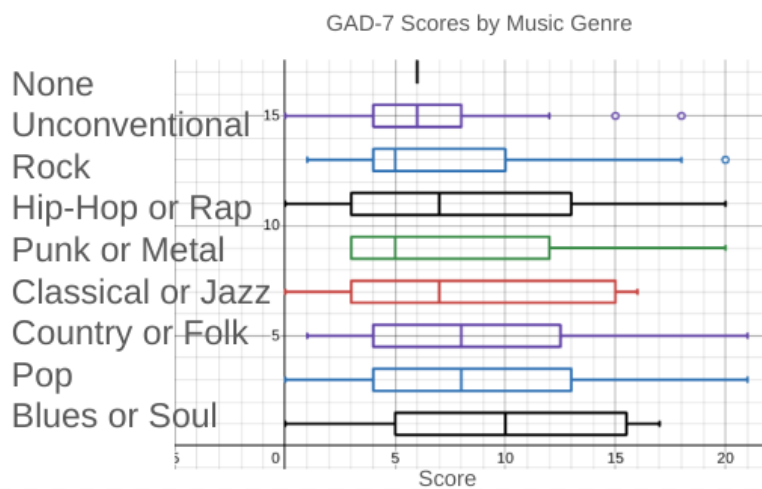
Histogram of GAD7 SCORE



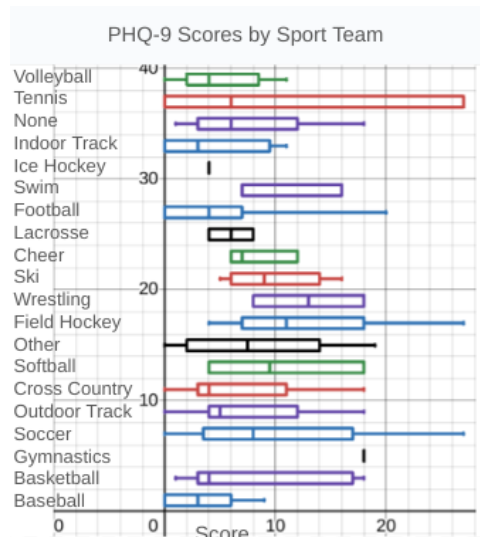
These two histograms display the PHQ-9 score and the GAD-7 score on the x-axis and the number of respondents on the y-axis. These graphs contain all Wachusett student respondents. The PHQ-9 scores had a mean of 8.144 with a standard deviation of 6.621. The GAD-7 scores had a mean of 8.092 with a standard deviation of 5.528. The shapes of the PHQ-9 and GAD-7 histograms appear to be unimodal and skewed to the right. There do not appear to be any outliers on either histogram. The means of both histograms lie in the “Mild” range of scores, which is 4.50-10.00. According to professional guidelines, the “Mild” range of scores refers to the stage at which respondents' symptoms should be watched over a few weeks to notice any increase or decrease in behavior. For the purposes of our survey, we will not be monitoring the respondents' symptoms or engaging in involvement with medical professionals; we will only observe trends in data.



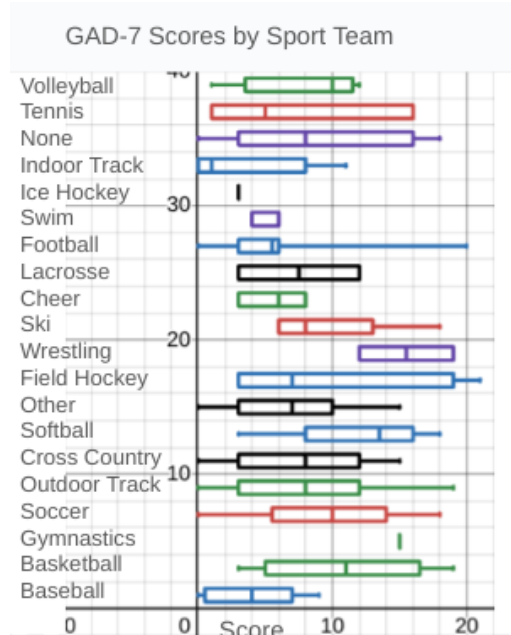
The survey data collected were organized into box plots of PHQ-9 scores split by music genre. Comparing each stacked plot, scores are generally similar across the different genres. This can be noticed by the significant overlap across all of the plots. They all have relatively similar IQRs, aside from “None,” due to only one person selecting that response. The genre “Country or Folk” is the only option with an outlier with a PHQ-9 score of 27, 22 points higher than the median for “Country or Folk.” The general distribution for each music genre appears to be skewed right, showing lower PHQ-9 scores.



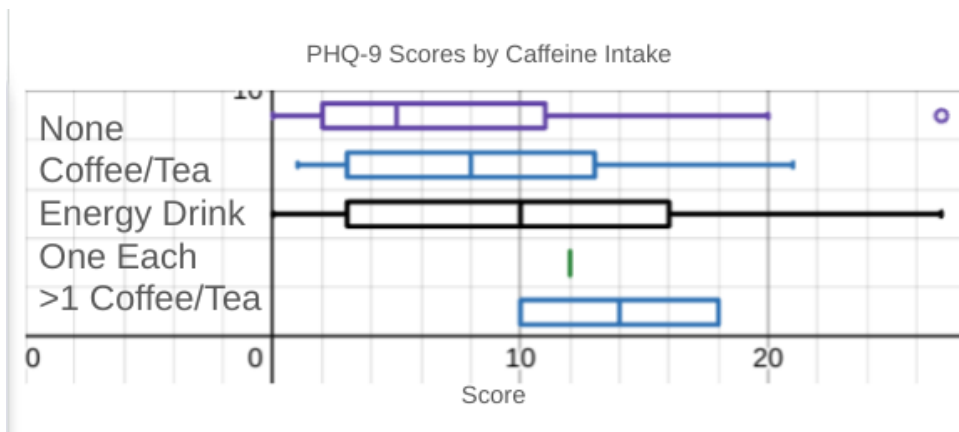
The box plots for GAD-7 compared to music genres show slightly different trends than the PHQ-9 compared to music genres. Similar to the PHQ-9 plots, most of the spreads are relatively alike, with “None” still being very small due to only one response; however, the IQRs for “Unconventional” and “Rock” are smaller than the others and have outliers. Likewise, in the PHQ-9 graph, “Blues or Soul” still has the highest median. The general distribution for each music genre appears to be skewed right, showing lower GAD-7 scores, except for “Unconventional,” which appears to be relatively symmetric.



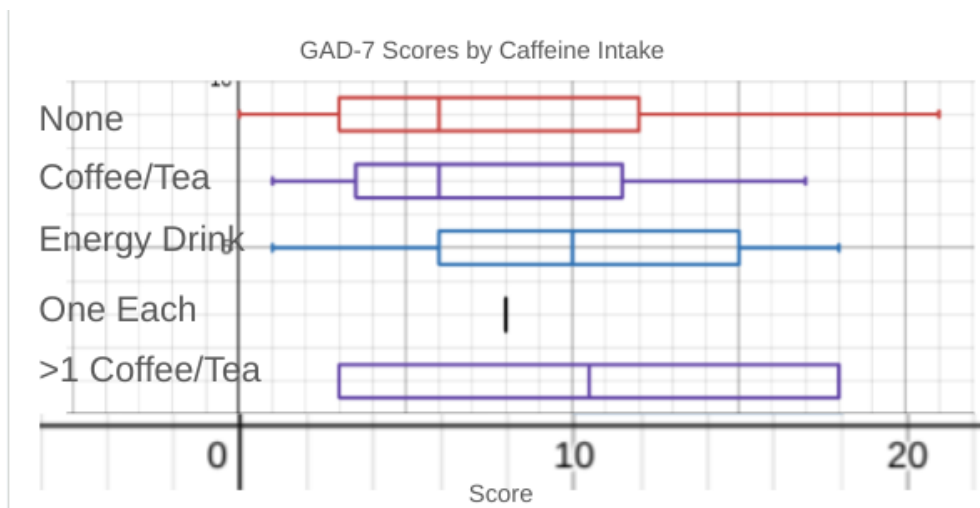
PHQ-9 scores compared to sports teams show various trends. Firstly, sports like “Gymnastics” and “Ice Hockey” had very few respondents, causing their boxplot spreads to be very small. Secondly, “Tennis” has a very large IQR, which means the middle 50% is spread across the whole range of responses. The plots overlap a lot, so a clear association cannot be seen. Interestingly, no sports team had an outlier, which shows that most of the data is within a similar range.



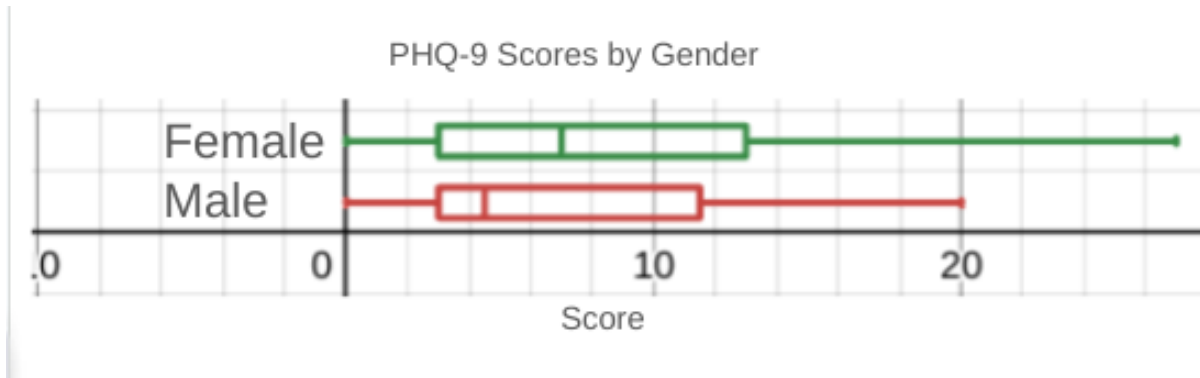
GAD-7 scores compared to sports teams have very similar trends to PHQ-9 compared to sports teams, again with much overlap. IQRs for both “Swim” and “Football” are much smaller than when compared to PHQ-9. Again, there are no outliers for any of the sports, so similar to the PHQ-9 scores, GAD-7 scores are generally within a certain range for each sport. Due to the lack of respondents who are gymnasts or play ice hockey, the boxplot ranges for those sports are very small.



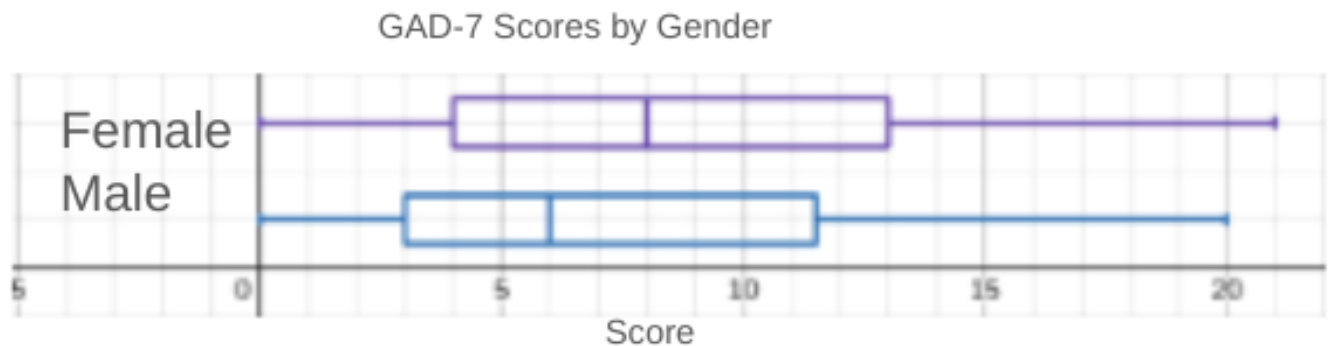
The boxplots for PHQ-9 scores compared to Caffeine Intake appear to show a trend. Higher PHQ-9 scores seem to be associated with higher caffeine intake. However, causation cannot be determined. The boxplot spread representing “One Each” is very small due to the lack of respondents picking that option. Therefore, there is not enough data to consider it. The option “None” has an outlier at a PHQ-9 score of 27, which could be accounted for by an underlying variable.



The boxplots for GAD-7 scores compared to Caffeine Intake appear to show a similar but less strong trend than the above graphs. The option “One Each” still has too few responses to be statistically significant. Medians for “None” and “Coffee/Tea” are equal, the median for “Energy Drink” is higher, and “>1 Coffee/Tea” is the highest. There is a lot of overlap, so the results are relatively similar. Lastly, the IQR of “>1 Coffee/Tea” overlaps with the other four IQRs. This shows that the spread of scores for each option is relatively similar.



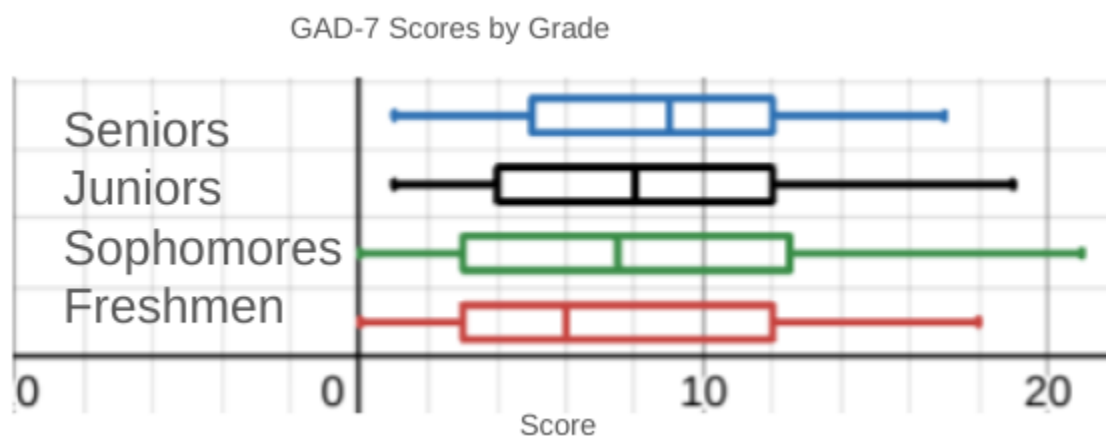
The boxplots for PHQ-9 scores compared to gender appear to show no real trends. There is a lot of overlap between the two spreads, which shows that they fall within a similar range. There appear to be no outliers in either plot. The median for “Female” appears slightly higher than “Male”. The IQRs are relatively the same size; however, “Female” has a higher maximum than “Male.”



The boxplots for GAD-7 scores compared to gender show similar results as the PHQ-9 plots. They again show a lot of overlap, suggesting that the scores fall within a certain range. The median for “Female” is slightly higher than “Male”. There appear to be no outliers in either plot.



The boxplots for PHQ-9 scores by grade level show relatively similar IQR spreads, with “Juniors” and “Sophomores” having the largest. All grades except “Juniors” appear skewed to the right, showing lower PHQ-9 score responses. There appear to be no outlier scores for either of the grade levels.



The four boxplots for GAD-7 scores by grade level all show a relatively similar-sized spread with unimodal or slightly skewed right distribution. There appear to be no outlier scores for either of the grade levels.

With all the data considered, no real conclusions can be drawn due to the overlap of the boxplots. Higher PHQ-9 scores appeared to be associated with more caffeine intake. We calculated the average PHQ-9 score (8.144) and standard deviation (6.621), and the average GAD-7 score (8.902) and standard deviation (5.527). For grade level, we observed that the grade levels' scores that fell above the mean for depression were Juniors and Seniors, and Seniors, Sophomores, and Juniors for anxiety. For gender scores, we found that females lay above the mean for both depression and anxiety. With caffeine intake, all categories but "None" fell above the mean for depression, and ">1 Coffee/Tea" and "Energy Drink" fell above the mean for anxiety. For sports, Field Hockey, Softball, Soccer, Ski, and Basketball players lay above the depression mean, while for Softball, Basketball, Field Hockey, Soccer, Ski, "None," and Outdoor Track players lay above the anxiety mean. Lastly, for music scores, Punk or Metal, Blues or Soul, Hip-Hop or Rap, Rock, and Pop listeners score above the depression mean, and Blues or Soul, Pop, Country or Folk, and Classical or Jazz listeners score above the anxiety mean. We can observe from this information that some characteristics are listed above the mean in both depression and anxiety. Yet, we cannot conclude any causation or imply that people with these traits will experience these mental health symptoms. The only group that potentially showed some association was PHQ-9 scores compared to Caffeine Intake. However, aside from that, the difference between medians and spread seemed to be too minuscule. A problem that we faced that made it harder to conclude was our small sample size. Boxplots were often too small because only one response for categories like "Wrestling" made the data difficult to analyze. Overall, we can conclude that there appears to be little to no association with the categories we tested for. However, we cannot assume any causation that links these characteristics to depression or anxiety just because there may be an association; we can only identify trends.

For future practice, we can suggest some improvements to run the questioning more smoothly and create better opportunities for data collection. We noticed in our proposal presentation and data collection that the "Sports" section of the survey confused some. We intended to see which sports, if any, had an association with depression and anxiety symptoms. However, this aim was confused with extracurriculars in general, as we received questions and responses like course selection and activities unrelated to sports like theatre. In our questionnaire, we could put a specifier explaining only to submit responses of which sports the respondents play in the "Other" selection, where they could input any answer. We could also categorize the sports into "Contact," "Non-Contact," and "Artistic" sports for the respondents to choose from. This would organize our data, giving us less specific data to analyze, which may allow us to see a clearer trend in which types of sports have an association. Another improvement could be to increase our sample size from 200 to 400 students. This would hopefully provide us with more responses. Only a fraction of our target sample of 200 students responded to the survey, leaving us with some responses with only one respondent representing the category. We could also provide more interesting questions about personal characteristics or concerns we were curious about to see if they had any association with depression and anxiety, like course level (AP, Honors, CPA, CP).